WORKED SOLUTION

STEP II 2017, Question 12

Probability and Statistics • Poisson Processes

STEP II 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q12

The key idea

Two independent Poisson streams of fish merge into a single Poisson stream whose rate is the sum of the two rates; that is what the convolution in (i) shows. For the timing questions, turn statements about *when* the first fish arrives into statements about *how many* fish have been caught: "Adam's first fish comes after time t" means exactly "Adam catches no fish in $[0,t]$", and the Poisson formula gives that probability straight away.

Solution

Throughout, X and Y are the numbers of fish caught by Adam and Eve in an interval of length T, so $X \sim \text{Po}(\lambda T)$ and $Y \sim \text{Po}(\mu T)$, independently.

Part (i)

For an integer $r \geq 0$, the event $X+Y=r$ splits into the disjoint events $\{X=j,\ Y=r-j\}$ for $j=0,1,\ldots,r$. Using independence,

$$\begin{aligned}\text{P}(X+Y=r) &= \sum_{j=0}^{r} \text{P}(X=j)\,\text{P}(Y=r-j) = \sum_{j=0}^{r} \frac{e^{-\lambda T}(\lambda T)^j}{j!}\cdot\frac{e^{-\mu T}(\mu T)^{r-j}}{(r-j)!} \\ &= \frac{e^{-(\lambda+\mu)T}}{r!}\sum_{j=0}^{r}\binom{r}{j}(\lambda T)^j(\mu T)^{r-j} = \frac{e^{-(\lambda+\mu)T}((\lambda+\mu)T)^r}{r!},\end{aligned}$$

where we multiplied and divided by $r!$ and then used the binomial theorem. This is the $\text{Po}((\lambda+\mu)T)$ probability, so the total catch in time T is Poisson with parameter $(\lambda+\mu)T$.

Part (ii)

For $0 \le r \le k$,

$$\mathrm{P}(X = r \mid X+Y = k) = \frac{\mathrm{P}(X = r,\ Y = k-r)}{\mathrm{P}(X+Y=k)} = \frac{\dfrac{e^{-\lambda T}(\lambda T)^r}{r!} \cdot \dfrac{e^{-\mu T}(\mu T)^{k-r}}{(k-r)!}}{\dfrac{e^{-(\lambda+\mu)T}((\lambda+\mu)T)^k}{k!}},$$

using part (i) for the denominator. The exponentials cancel, and so do the powers of T (there are $r + (k-r) = k$ on top and k underneath), leaving

$$\mathrm{P}(X = r \mid X+Y = k) = \binom{k}{r}\left(\frac{\lambda}{\lambda+\mu}\right)^r\left(\frac{\mu}{\lambda+\mu}\right)^{k-r}.$$

So, given a total of k, Adam's catch is $\mathrm{B}(k,\ \frac{\lambda}{\lambda+\mu})$. Each of the k fish is, in effect, independently Adam's with probability $\frac{\lambda}{\lambda+\mu}$. Notice that T has disappeared.

Part (iii)

Let A and E be the times, measured from the start, at which Adam and Eve catch their first fish. Adam's first fish comes after time t exactly when he catches nothing in $[0, t]$, so

$$\mathrm{P}(A > t) = \mathrm{P}(\mathrm{Po}(\lambda t) = 0) = e^{-\lambda t}, \qquad \text{and similarly} \qquad \mathrm{P}(E > t) = e^{-\mu t}.$$

Hence A has probability density function $\lambda e^{-\lambda t}$ for $t \ge 0$ (differentiate $1 - e^{-\lambda t}$), and A and E are independent because the two catches are.

? **Ask yourself** How do we combine a density for A with a probability for E?

Condition on the value of A: if Adam's first fish comes at time t, he is first precisely when $E > t$, which has probability $e^{-\mu t}$ by independence. Summing over all t,

$$\mathrm{P}(A < E) = \int_0^\infty \lambda e^{-\lambda t}\, e^{-\mu t}\, \mathrm{d}t = \lambda\left[\frac{-e^{-(\lambda+\mu)t}}{\lambda+\mu}\right]_0^\infty = \frac{\lambda}{\lambda+\mu}.$$

(Ties have probability zero, since E has a continuous distribution.) So the first fish is Adam's with probability $\dfrac{\lambda}{\lambda+\mu}$, the same as the binomial parameter in (ii).

Part (iv)

We want $\mathrm{E}(M)$, where $M = \max(A, E)$ is the time by which both have caught a fish. Write $m = \min(A, E)$, the time of the first fish caught by either of them. For any two numbers, max + min is their sum, so

$$M = A + E - m \qquad \Longrightarrow \qquad \mathrm{E}(M) = \mathrm{E}(A) + \mathrm{E}(E) - \mathrm{E}(m).$$

$\mathrm{E}(A)$ *and* $\mathrm{E}(E)$. From the start, Adam's catches form the same kind of process as from the moment of any catch, so the expected wait for his first fish is the same as the given expected gap, $\mathrm{E}(A) = \lambda^{-1}$, and likewise $\mathrm{E}(E) = \mu^{-1}$. (Directly: $\mathrm{E}(A) = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t = \lambda^{-1}$, integrating by parts.)

Ask yourself What kind of process is "the first fish caught by either of them"?

By part (i), the combined catch in any interval of length t is $\mathrm{Po}((\lambda+\mu)t)$. So the combined catch is exactly the same kind of process as Adam's, but with rate $\lambda+\mu$ in place of λ, and its expected waiting time to the first fish is $\mathrm{E}(m) = (\lambda+\mu)^{-1}$. (Directly: $\mathrm{P}(m > t) = \mathrm{P}(A > t)\,\mathrm{P}(E > t) = e^{-(\lambda+\mu)t}$.) Therefore

$$\mathrm{E}(M) = \frac{1}{\lambda} + \frac{1}{\mu} - \frac{1}{\lambda+\mu} = \frac{\lambda^2+\lambda\mu+\mu^2}{\lambda\mu(\lambda+\mu)}.$$

Checks

- If $\lambda = \mu$, part (iii) gives $\frac{1}{2}$, as symmetry demands, and part (iv) gives $\frac{3}{2\lambda}$: wait $\frac{1}{2\lambda}$ for the first fish, then $\frac{1}{\lambda}$ more for the other person, by the same "fresh start" reasoning.
- If $\mu \to \infty$ (Eve catches fish instantly), part (iv) tends to $\frac{1}{\lambda}$, the time Adam needs on his own.
- The answer to (iv) is at least $\max(\frac{1}{\lambda}, \frac{1}{\mu})$, as it must be, since $\frac{1}{\mu} > \frac{1}{\lambda+\mu}$.
- With $\lambda = 1.3$, $\mu = 0.7$, the formulas give 0.65 and 1.6978; a computer simulation of 2×10^5 fishing trips gave 0.651 and 1.700.

Common slips

- In (iv), adding $\frac{1}{\lambda} + \frac{1}{\mu}$, or taking $\max(\frac{1}{\lambda}, \frac{1}{\mu})$: the expectation of a maximum is neither of these.
- In (ii), forgetting to divide by $\mathrm{P}(X+Y=k)$, which is what turns the joint probability into a conditional one.

Beyond STEP

What you have met here is the *Poisson process*: waiting times between events are exponential and memoryless, independent Poisson processes superpose into another one, and each event is independently labelled with probability proportional to its rate (the reverse operation, "thinning", also works). These facts are the foundation of queueing theory and continuous-time Markov chains. Grimmett and Welsh, *Probability: An Introduction*, treats the Poisson process and exponential distribution in a way that follows on naturally from this question.

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